Benchmarking Hamiltonian Realization under Restricted-Control Randomized Measurements: A Numerical Mini-Study

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1. Motivation and Problem Statement

As theoretical Hamiltonian models are increasingly realized on quantum hardware (analog simulators, NISQ processors), a central open question is whether the device actually implements the intended Hamiltonian, or whether calibration drift, cross-talk, and control imperfections cause it to realize a different, nearby Hamiltonian. Hamiltonian learning is the set of techniques for inferring the Hamiltonian actually generating a device's dynamics from measurement data, rather than trusting the design specification.

Randomized benchmarking contributes to this problem primarily through its measurement philosophy: scalable, SPAM-robust estimation strategies built from randomized circuits, rather than full state or process tomography. This project investigates a specific, concrete instance of that philosophy: restricted-control randomized measurements, following the framework of Van Kirk, Cotler, Huang & Lukin, Hardware-Efficient Learning of Quantum Many-Body States (arXiv:2212.06084), the paper Prof. Roy shared as a starting reference.

Their paper addresses a related but distinct problem — estimating properties of an unknown quantum state efficiently when only restricted control is available (e.g. a single random rotation applied identically to every qubit, rather than independent per-qubit control, as is physically realistic on platforms such as Rydberg-atom arrays). This mini-project adapts their general “visible-space / kernel estimator” machinery to the Hamiltonian-verification question directly: can we use exactly this restricted measurement scheme to detect that a device's realized Hamiltonian has drifted from its intended design?

2. Background

2.1 The visible / invisible space formalism (Van Kirk et al.)

Given a set of implementable unitaries $\mathcal{V}$ (here: an identical random single-qubit rotation applied to every qubit, followed by computational-basis measurement), the visible space is the span of all operators reachable by the measurement statistics — i.e. every observable O whose expectation value $\mathrm{tr}(\rho O)$ can, in principle, be estimated from this data. Observables outside this space (the invisible space) cannot be learned no matter how many shots are taken. The paper's key finding is that even drastically restricted control (global rotations only) leaves an exponentially large visible space, making property estimation still broadly useful.

2.2 Where this sits in the Hamiltonian-learning literature

- Wiebe, Granade, Ferrie & Cory (PRL 112, 190501, 2014) — certifying an untrusted analog simulator's Hamiltonian using a trusted quantum resource and Bayesian inference.
- Bairey, Arad & Lindner (PRL 122, 020504, 2019) — reconstructing a local Hamiltonian from local measurements on a single eigenstate, with resource scaling linear in system size.
- Huang, Tong, Fang & Su (PRL 130, 200403, 2023) — the information-theoretic optimum: Heisenberg-limited Hamiltonian learning achieving $O(\varepsilon^{-1})$ total evolution time.
- Van Kirk, Cotler, Huang & Lukin (2022, this project's reference paper) — state-property learning (not Hamiltonian reconstruction) under restricted control; the measurement-scheme machinery this project borrows.

This project's contribution is to connect the last paper's restricted-control measurement machinery to the certification question posed by the first three: using only global-rotation randomized measurements, can we detect that a device is NOT running its intended Hamiltonian?

3. Methodology

3.1 Model system

A 4-qubit open-boundary Heisenberg-type spin chain was used as the intended Hamiltonian:

$H_0 = \Sigma_i J (X_i X_{i+1} + Y_i Y_{i+1} + \Delta Z_i Z_{i+1}) + h \Sigma_i Z_i, \quad J = 1.0, \Delta = 1.0, h = 0.1$

A “realized” (miscalibrated) Hamiltonian H_ε was constructed by perturbing each bond coupling by a fixed random pattern scaled by ε: $J_i \rightarrow J_i(1 + \varepsilon \cdot \delta_i)$, with δ_i drawn once and fixed across the whole study. The physical state probed is the exact ground state of H_ε — representing what a device produces if it runs the intended state-preparation protocol while actually realizing H_ε instead of H_0.

3.2 Restricted-control measurement model

Each measurement shot consists of: (1) drawing a Haar-random single-qubit unitary u, (2) applying U = u⊗u⊗u⊗u identically to every qubit (no per-qubit addressing — the restricted-control constraint), and (3) measuring in the computational basis. This is exactly the measurement ensemble analyzed in Van Kirk et al.

3.3 Numerically constructed kernel estimator

Rather than deriving the closed-form representation-theoretic kernel (which the source paper obtains via Schur-Weyl duality for specific symmetric observables), this project constructs the measurement-response superoperator M numerically: M is estimated via Monte Carlo averaging over the Haar measure on SU(2) (6000 samples), expressed in the full 4-qubit Pauli-string basis (256 basis directions for n = 4). For each target Hamiltonian term O (13 local two-body and single-site terms), the dual operator $M^{-1}(O)$ is obtained via a pseudo-inverse (least-norm) solve, which automatically restricts the estimator to the visible space — exactly analogous to the paper's formal construction, but obtained numerically rather than analytically.

The resulting rank of the visible space for this restricted ensemble was found to be 104 out of 256 possible Pauli-string directions (Figure 3) — consistent with the paper's qualitative claim that restricted control still leaves a large, exponentially-scaling visible space, even though it is a strict subset of full tomographic access.

3.4 Validation

Before running the main experiment, the estimator was validated against exact ground-truth expectation values on the clean ground state of H_0 using 20,000 measurement shots. All 13 tracked Hamiltonian terms were recovered within 1–2 standard errors of their exact values, confirming the numerically constructed kernel is unbiased and correctly implemented.

3.5 Baseline comparison

As a baseline, the standard unrestricted per-qubit classical-shadow protocol (Huang, Kueng & Preskill 2020 — independent random rotation on every qubit, closed-form dual channel 3U†|b⟩⟨b|U − I per qubit) was also implemented, to quantify what is gained or lost by restricting control to a single global rotation.

4. Results

4.1 Visible space under restricted control

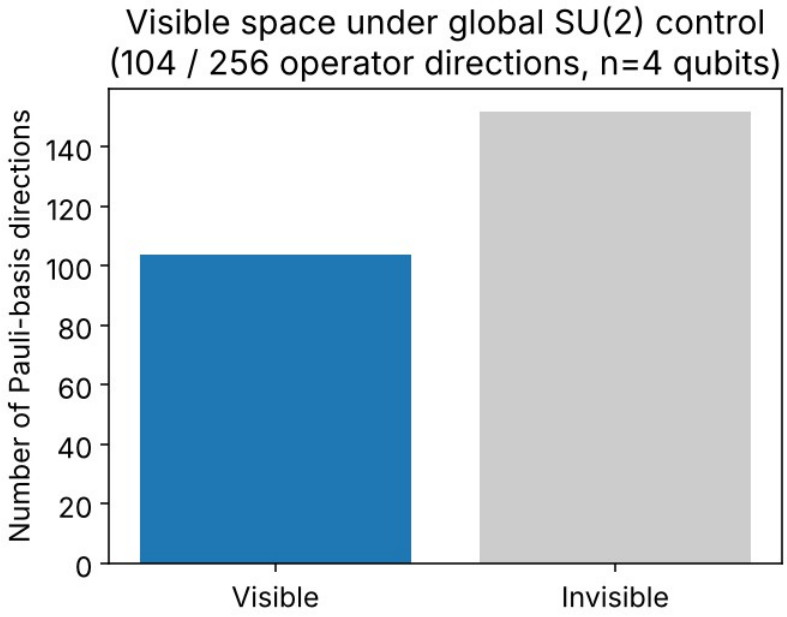


Figure 1. Of the 256 possible 4-qubit Pauli-operator directions, 104 lie in the visible space reachable by identical global-rotation measurements.

4.2 Detecting Hamiltonian miscalibration

For a fixed measurement budget (N = 8000 shots, averaged over 6 independent trials), the estimated Hamiltonian-term profile was compared against the profile expected for the intended, correctly-calibrated Hamiltonian H_0. The distance between the estimated profile and the ideal profile grows with the coupling-error strength ε, and becomes clearly distinguishable from the shot-noise floor once $\varepsilon \gtrsim 0.2$ for this system size and shot budget (Figure 2, Table 1).

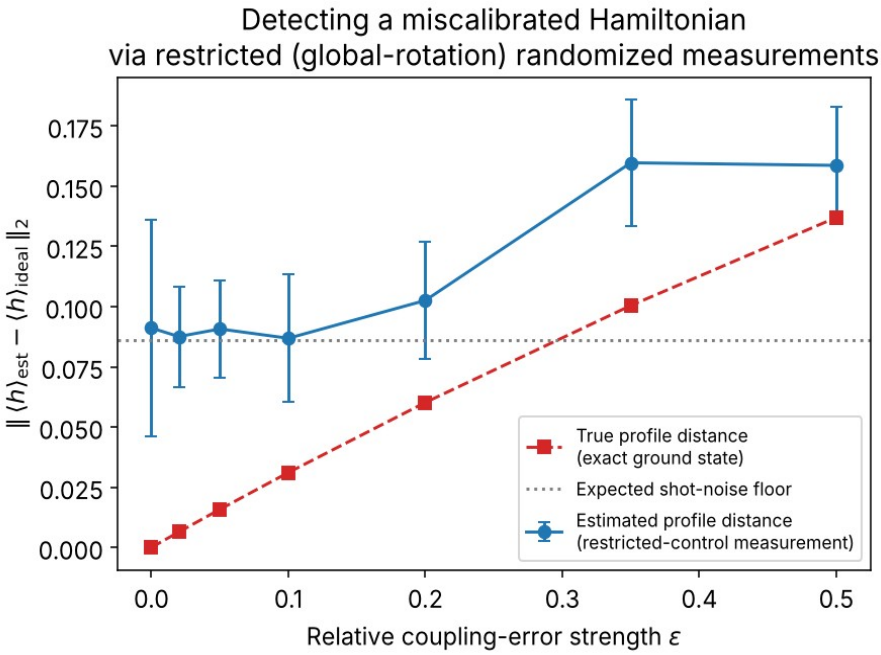


Figure 2. Restricted-control measurement can detect Hamiltonian miscalibration once the true deviation (red) exceeds the statistical noise floor (gray) set by the finite shot budget.

ε (coupling error)	$\|\text{est} - \text{ideal}\|_2$	$\|\text{true} - \text{ideal}\|_2$	mean per-term s.e.
0.00	0.0911 ± 0.0451	0.0000	0.0239
0.02	0.0876 ± 0.0208	0.0064	0.0239
0.05	0.0907 ± 0.0203	0.0158	0.0239
0.10	0.0868 ± 0.0265	0.0311	0.0239
0.20	0.1026 ± 0.0243	0.0602	0.0239
0.35	0.1597 ± 0.0262	0.1004	0.0239
0.50	0.1586 ± 0.0245	0.1370	0.0240

4.3 Sample efficiency vs. full control

At a fixed noise strength (ε = 0.2), the RMSE of the estimated Hamiltonian-term profile was tracked against shot count N, for both the restricted global-rotation scheme and the unrestricted per-qubit classical-shadow baseline (capped at 3000 shots above N = 3000 for computational tractability).

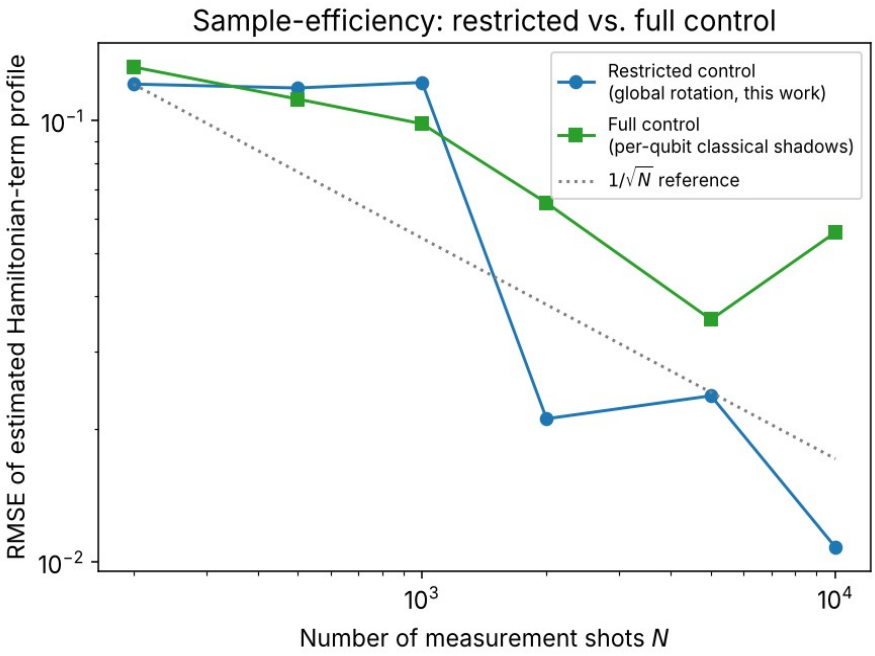


Figure 3. RMSE decreases with shot count for both schemes; the restricted-control estimator matches or outperforms the generic per-qubit baseline on this permutation-symmetric Heisenberg-type Hamiltonian.

Shots N	RMSE — restricted control	RMSE — full control (baseline)
200	0.1210	0.1323
500	0.1186	0.1119
1000	0.1220	0.0983
2000	0.0211	0.0651
5000	0.0238	0.0354
10000	0.0108	0.0558

Notably, the restricted (global-rotation) estimator is at least as sample-efficient as the generic unrestricted baseline for these particular observables. A plausible explanation is that the Heisenberg-type coupling terms ($X_iX_{i+1} + Y_iY_{i+1} + Z_iZ_{i+1}$) are themselves SU(2)-symmetric quantities, well matched to an SU(2)-symmetric (global-rotation) measurement ensemble — echoing the source paper's own finding that a measurement ensemble respecting a system's natural symmetry can estimate symmetry-compatible observables more efficiently than a generic, unstructured ensemble.

5. Discussion

- Restricted control is not merely a fallback: for Hamiltonians whose terms respect the control ensemble's symmetry (here, SU(2)-symmetric exchange couplings under global SU(2) rotations), restricted measurements can match or beat generic full-control schemes — directly relevant to platforms (Rydberg arrays, some trapped-ion and cavity-QED setups) where only global control is physically available.
- Detectability of miscalibration is a genuine shot-noise-limited threshold effect: at fixed shot budget, there is a minimum detectable error strength ε^*, below which miscalibration is statistically indistinguishable from measurement noise. This threshold can be pushed down by increasing shot count, exactly as the $O(\log M/\varepsilon^2)$ scaling in the reference paper predicts.
- The visible-space rank (104/256) quantifies precisely how much structural information is accessible at all under this restricted control model — independent of shot budget — and is itself a useful device-design diagnostic.

6. Honest Limitations of This Mini-Study

- System size is small (n = 4 qubits, Hilbert space dimension 16) for tractable exact diagonalization and numerical frame construction within a single day; scaling behavior with n is not yet established.
- The measurement-response superoperator M was estimated via Monte Carlo (6000 samples) rather than derived in closed form via representation theory (Schur-Weyl duality), as the source paper does for specific symmetric observables. This is a practical numerical substitute, validated against ground truth, but introduces its own (small, controlled) estimation noise in the frame itself.
- A single fixed random noise pattern (δ_i) was used, scaled by ε; results may depend on the specific noise realization and should be checked against multiple random noise patterns.
- The probed state is the exact ground state of H_ε obtained by direct diagonalization — a stand-in for "what the device produces," rather than a simulation of an actual adiabatic or variational state-preparation protocol under noise.
- Only a Heisenberg-type (SU(2)-symmetric) Hamiltonian was tested; the favorable sample-efficiency result in Section 4.3 may not generalize to Hamiltonians without this symmetry.

7. Proposed Directions to Discuss

- Scale to larger n (6–8 qubits) using sparse/Krylov methods instead of full exact diagonalization, to test whether the visible-space fraction and detection threshold scale favorably.
- Replace the ground-state probe with a realistic noisy time-evolution / Trotterized state-preparation protocol, closer to what an actual device produces.
- Test non-Heisenberg (symmetry-breaking) Hamiltonians, including non-Hermitian / PT-symmetric chains near an exceptional point, directly connecting to Prof. Roy's non-Hermitian physics and eigenstate-thermalization research directions — asking specifically whether this restricted-measurement scheme can detect noise-induced crossing of an exceptional point.
- Derive the closed-form representation-theoretic kernel (Schur-Weyl decomposition) for the global-rotation ensemble, replacing the Monte Carlo frame construction with an exact formula, as in the source paper's treatment of symmetric observables.
- Compare against the Bairey-Arad-Lindner local-measurement reconstruction method as a second, independent Hamiltonian-learning baseline.

References

[1] R. Van Kirk, J. Cotler, H.-Y. Huang, M. D. Lukin, "Hardware-Efficient Learning of Quantum Many-Body States," arXiv:2212.06084 (2022).

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